

Source Code – GenAI in Testing

package.json

```
{
  "name": "genai-kyc-aml-testing",
  "version": "1.0.0",
  "description": "KYC and AML Testing Framework",
  "scripts": {
    "test": "jest"
  },
  "dependencies": {
    "@faker-js/faker": "^9.0.0"
  },
  "devDependencies": {
    "jest": "^30.0.0"
  }
}
```

kycDataGenerator.js

```
const { faker } = require('@faker-js/faker');

function generateKYCData() {
  return {
    customerId: faker.string.uuid(),
    fullName: faker.person.fullName(),
    email: faker.internet.email(),
    mobile: faker.phone.number(),
    address: faker.location.streetAddress(),
    dob: faker.date.birthdate(),
    pan: "ABCDE1234F",
    aadhaar: "123456789012"
  };
}

for(let i=0;i<10;i++){
  console.log(generateKYCData());
}
```

amlDataGenerator.js

```
const { faker } = require('@faker-js/faker');

function generateAMLTransaction() {
  return {
    transactionId: faker.string.uuid(),
    customerId: faker.string.uuid(),
    amount: faker.finance.amount(1000, 1000000),
    currency: "INR",
    country: faker.location.country(),
    riskFlag: faker.datatype.boolean()
  };
}

for(let i=0;i<10;i++){
  console.log(generateAMLTransaction());
}
```

testScenarios.js

```
const testScenarios = [
  {
    scenarioId: "TS001",
    description: "Verify customer KYC registration"
  },
  {
    scenarioId: "TS002",
    description: "Verify PAN validation"
  },
  {
    scenarioId: "TS003",
    description: "Verify AML suspicious transaction detection"
  },
  {
    scenarioId: "TS004",
    description: "Verify chatbot response accuracy"
  }
];

console.table(testScenarios);
```

testCases.js

```
const testCases = [
  {
    testCaseId: "TC001",
    scenario: "KYC Registration",
    expectedResult: "Customer registered successfully"
  },
  {
    testCaseId: "TC002",
    scenario: "PAN Validation",
    expectedResult: "PAN accepted"
  },
  {
    testCaseId: "TC003",
    scenario: "AML Detection",
    expectedResult: "Suspicious transaction flagged"
  }
];

console.table(testCases);
```

rtmGenerator.js

```
const rtm = [
  {
    requirementId: "REQ001",
    userStoryId: "US001",
    testCaseId: "TC001",
    status: "Passed"
  },
  {
    requirementId: "REQ002",
    userStoryId: "US002",
    testCaseId: "TC002",
    status: "Passed"
  },
  {
    requirementId: "REQ003",
    userStoryId: "US003",
    testCaseId: "TC003",
    status: "Passed"
  }
]
```

```
];  
  
console.table(rtm);
```

kyc.test.js

```
describe('KYC Validation Tests', () => {  
  
  test('Customer name should not be empty', () => {  
    const customerName = 'John Smith';  
    expect(customerName).not.toBe('');  
  });  
  
  test('PAN should contain 10 characters', () => {  
    const pan = 'ABCDE1234F';  
    expect(pan.length).toBe(10);  
  });  
  
});
```

aml.test.js

```
describe('AML Monitoring Tests', () => {  
  
  test('High value transaction should trigger review', () => {  
    const amount = 1500000;  
    expect(amount).toBeGreaterThan(1000000);  
  });  
  
  test('Risk flag should be true for suspicious transaction', () => {  
    const riskFlag = true;  
    expect(riskFlag).toBe(true);  
  });  
  
});
```

Jest Execution Command

```
npm install  
npm test
```

Expected Output

```
PASS ./kyc.test.js  
PASS ./aml.test.js  
  
Test Suites: 2 passed, 2 total  
Tests: 4 passed, 4 total
```